Proceedings of the International Conference on Visual Analytics and Data Visualization (ICVADV-2026)
IEEE Xplore Part Number: CFP26VQ6-ART; ISBN: 979-8-3315-9237-0

A Feature-Optimized PLSR Framework for Non-Destructive Assessment of Mango Pulp Firmness using Vis/NIR Hyperspectral Imaging

S.MEENAKSHI
Department of Electronics and Communication
Chennai Institute of Technology
Chennai, India
meenakshis@citchennai.net

P.SURESH KUMAR
Department of Electronics and Communication
Chennai Institute of Technology
Chennai, India
sureshkumarp@citchennai.net

SRIKAR.S
Department of Electronics and Communication
Chennai Institute of Technology
Chennai, India

Abstract: Mango is a tropical fruit significantly supports global economy for its rich flavor, nutrition and aroma. The customer acceptance of the fruit depends on color, internal and external quality attributes of the fruit. Nearly 1/5th of the mango fruit quality reduced during ripening due to drastic changes in the physiochemical activities. A non-destructive visNIRS and spectral values captured with the wavelength from 900 to 1700 for real-time detection of pulp firmness measurement. The proposed PLSR+CARS+SVR model offers an innovative solution in non-destructive testing technological frameworks. The feature vectors are iteratively extracted using hybrid CARS+SVR model and gives a quantitative result for Root Mean Square Error (RMSE) as 0.21 and correlation coefficient R^2 as 0.95. This method considerably reduces the food wastage and improvises postharvest management.

Keywords- Pulp firmness, Partial Least Squares Regression (PLSR), Competitive Adaptive Reweighted Sampling (CARS) Support Vector Regressor

Introduction

Mango (Mangifera indica L.) known as King of fruits holding significant position in global market for its nutrition benefits, aroma and economic value. The fruit development process is divided as cell division, expansion and maturation. The duration of these activities has been depicted in Figure-I..

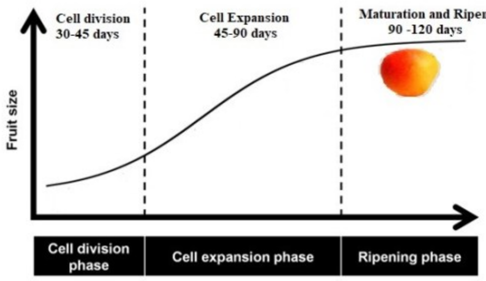

Figure-I Mango fruit development Stages

The cell division stage is responsible for the determination of final fruit size, the fruit expansion can be determined from cell enlargement and conversion of starch to sugar, colour change and harvest readiness is found from maturation stage

Understanding the biological processes of mango fruit development is essential for determining the correct harvest timing, ensuring maximum fruit quality, storability, and consumer satisfaction. The main factors determine the quality of mango fruit includes pulp colour, firmness and aroma. Physiological maturity is more important to withstand to survive in long journeys, controlled temperature and rough handling. Premature fruits lacks with aroma, causes shrivelling whereas overripe fruit have poor shelf life [1].

The mango pulp contains micronutrients, fibers and vitamins and constitutes 40-60% of fruit weight. The standardized range of pulp includes14–18°Brix, acidity value is 4–6% and vitamin C range is 18g/kg [2]. The typical values of fruit maturity parameters has been depicted in Table-I. Among the factors pulp firmness value constant during growth period and gets decreased around 5% after reaching maturity and 0.3N considered as threshold for consumption. The firmness has direct correlation with TSS and peel thickness. It directly reflects maturity, shelf life, consumer acceptability and post-harvest quality. Normally maximum and minimum firmness observed in the top and bottom position of the fruit.

Even with the development of innovative processing methods maintaining the quality of the mango fruit is a challenge due to its volatile and sensitive nature. Traditional destructive physical and chemical analysis provides accurate results but not scalable. Inspections with human error cause inconsistencies in grading.

Table I Fruit Maturity Indicators and their typical Ranges

Indicator	Description	Typical Range / Observation
Peel Color	Chlorophyll degradation	Green → Light Yellow / Orange-Red (variety dependent)
Pulp Firmness	Softens with enzyme activity	Reduces gradually during maturity
Specific Gravity	Mature fruit sinks in water	>1.0 (sink) = mature
Total Soluble Solids (TSS)	Conversion of starch to sugars (°Brix)	12–20° Brix (variety dependent)

Destructive testing and human errors needs to be replaced to reduce commercial losses and to increase speed of quality detection. Non-destructive testing (NDT) evaluates the quality based on properties of material without causing any internal damage. The detection of firmness with nondestructive techniques works based on electrical, spectral, optical ultrasound and acoustic properties. Among acoustic technique have good correlation with fruit texture properties. Other parameters include density, mass, cell wall thickness, juiciness etc., mass, and shape.

The comparison of the techniques such as Universal Testing Machine and low-mass impact tester for the determination of mature to over mature stage of mango fruit from two different cultivators has been illustrated. It shows that the firmness doesn't varied in immature mature stage and deteriorates during fully ripen stage. Also the method shows same result for different cultivators. Comparing the speed the impact method works rapidly and accurately [3]. The pulp firmness has even ripening at the top, middle, and bottom positions of the fruit on harvest day and the seventh day of storage, according to a firmness determination using the Di Texture analyzer [4]. The stiffness then decreased to 0.3 to 3.5 N. The pulp firmness reduced by roughly 5% as the storage period increased, and 0.3 N is regarded as the consumption threshold.

A non-destructive texture analyzer with aluminum probe of 35cm diameter and 1mm space for placing mangos to analyze the firmness individual basis has been explored. From the measured values a Non-linear indexed regression using statistical package of R programming is applied to measure the variation. The results shows titratable acidity has high position correlation [5]. The physicochemical properties and properties related to firmness like soluble solids content (SSC), vitamin content and titratable acidity (TA) has been explored [6] with visible and near-infrared (VNIR) spectroscopy. The feasibility of this method widely accepted for determining chemical components and for some parameters the method provides uncertain results.

The properties of an acoustic vibration sensor have been used to address the hardness of mango fruit. This uses the NdYAG pulsed laser-induced plasma shock technique to create a Rayleigh wave. From that values observed that the wave propagation velocity, density and Poisson's ratio have correlation with firmness. The changes produced due to wave propagation velocity are large compared with other parameters and decreased to 4% in 96 hours [7].

Detection of insects in mango pulp from audio frequency extraction with noninvasive acoustic sensors using machine learning has been discussed. Among the different acoustic sensor MSME sensor selected as best and placed in soundproof chamber to minimize noise. The recorded audio signal is trained and classified with SVM classifier and provides an accuracy of 89.81% [8].

Application of ultrasound waves for testing solid cavity phenomenon releases, accumulated energy in the waves by alternating compression and decompression based on solid surface strength. Testing mango firmness with ultrasound waves and regression models is obtained to detect antioxidant capacity, ethanol and sonication time [9].This testing improves the accuracy and reduces the processing time.

The use of laser Doppler Vibrometry (LDV) and visNIRS for the non-destructive detection of mango fruit has been demonstrated[10]. The study determines ripeness quality of a fruit with firmness, sweetness and color. The method provides better results for total soluable content with poor predictions on firmness. To reduce the limitation of fruit size in acoustic vibration prediction technique a pneumatic-electromagnetic-driven impact force control with embedded accelerometer has been used for testing 156 mango fruits. A partial least squares regression based on competitive adaptive reweighted sampling (CARS-PLSR) statistical analysis is applied on the measured data and predicts that vibration signal has good correlation with mango firmness[11].
.

Accurate identification of fruit varieties based on texture, taste, vitamin C and Titratable Acidity (TA) content is determined with two distinct Near-Infrared Spectroscopy (NIRS) for four mango varieties. The various pre-processing techniques have improved the fruit's spectral quality, and Principal Component Analysis (PCA) is utilized to reduce dimensionality and streamline the spectral data.. To improve the predicting performance Synthetic Minority Over-sampling Technique (SMOTE) is applied and the varieties classified with Machine Learning (ML) models. The proposed method provides an accuracy of 95.0% for Vitamin C and 100% in mango variety classification [12].

VisNIRS models highly sensitive to temperature and provides information only in a single point on the fruit. However the hyper spectral models works on 2D image covers most surface area and provides spatial and structural information about each pixel with additional time and resources [13].Detection of peak values in a HSI is difficult due to high noise interference. Combination of spectroscopy with HSI images provides good reproducible results.

The major contributions of the work includes the acquisition of NIRS an spectral images, preprocessing dimensionality reduction , feature selection and extraction using CARS and Partial Least Squares Regression(PLSR) and nonlinear modeling with Support vector Regression(SVR) to assist quality inspection and post-harvest supply chain management.

I. MATERIALS AND METHODS

The dataset contains 186 images of four varieties of mango from different cultivators [14] in the Near Infrared Region (900 to 1700 nm). The information captured information has both desirable and unwanted data termed as noise [15]. This may be triggered by background noise, temperature fluctuations, uneven spectra, or interference from light scattering from the materials.

An image masking algorithm is applied to separate individual fruits from a bunch of fruits. The dataflow of the proposed method is depicted in Figure –II.As the captured data size is very large some dimensionality reduction techniques need to be applied after preprocessing. The preprocessing reduces the noise, drift and scattering in an image. Normally applied techniques are smoothing, transformation and scatter correlation. After preprocessing dimensionality reduction techniques are applied to reduce the redundancy, computational complexity and over fitting.

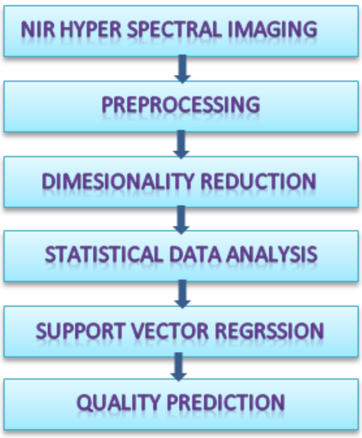

Figure-II Proposed Flow

The NIR images capture the content which is directly related with firmness such as water, sugar and pectin. The captured spectral data normally modelled with linear or nonlinear regression techniques to estimate the internal quality. As the spectral data is highly correlated traditional regression methods produces ineffective results.

```
# Input: NIR spectra (X), firmness reference values (Y)
# Output: Optimized PLSR model for mango pulp firmness
Initialization
Initialize Noruns # number of Monte Carlo runs
Initialize decay_rate, variable_list = all_wavelengths
best_RMSECV = infinity
best_variables = None

for k in range(1, Noruns + 1):

  # Step 1: Random Sampling
  X_train, Y_train = random_subset(X, Y, ratio=0.8)

  # Step 2: Build PLSR model
  model = PLSR(X_train[:, variable_list], Y_train)
  coeffs = abs(model.regression_coefficients)

  # Step 3: Exponential decay for variable elimination
  retain_count = int(len(variable_list) * np.exp(-decay_rate * k))

  # Step 4: Adaptive Reweighting
  sorted_idx = np.argsort(coeffs)[::-1]
  variable_list = variable_list[sorted_idx[:retain_count]]

  # Step 5: Evaluate model with cross-validation
  RMSECV = cross_validate(model, X_train[:, variable_list], Y_train

 if RMSECV < best_RMSECV:
    best_RMSECV = RMSECV
    best_variables = variable_list.copy()

 # Step 6: Build final model using best variables
 final_model = PLSR(X[:, best_variables], Y)
```

Among Partial Least Squares Regression (PLSR) is mostly preferred as it has the ability to handle complex data, capable of accommodating variations in fruit maturity and firmness changes. The pseudo code for detecting firmness value from NIR spectra is depicted as below.For optimization of PLSR k-fold cross validation is applied. PLSR effectively handles linearity among multiple variables with high computational efficiency.

The final regression coefficients are calculated and given as input to Competitive Adaptive Reweighted Sampling (CARS) for further reduction in redundant spectral information and finding group of dominant variable. The importance of each variable is found in iterative manner using Monte Carlo sampling, and on each run fixed number of variables is selected. For further refinement adaptive scaling and exponentially decreasing function (EDF) is applied in two step to select key variables [16]. Wavelength points with lower weights are eliminated using adaptive scaling, and variables with the lowest root mean square error (RMSECV) are chosen from cross validation. The application of exponential decay function minimizes the variables used and Monte Carlo sampling improves the model stability. The RMSECV value can be calculated from the actual sample y_i and predicted sample after cross validation y_{icv}.

$$RMSECV = \sqrt{\frac{\sum_{i=1}^{n}(y_i - y_{icv})^2}{n}}$$

where the number of samples denoted as n.

Further the nonlinearity among different features has been resolved with Support Vector Regression (SVR) by identifying more informative wavelength. It projects the selected spectral features through nonlinear relationships with strong generalization capability. The performance of the model is optimized by fine tuning kernel coefficient and evaluated using coefficient of determination (R^2) .It can be calculated from the predicted y_{ip} , actual and $\bar{y}$ average of the input values.

$$R^2 = \frac{\sum_{i=1}^{n}(y_{ip} - \bar{y})^2}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The predicted R^2 value can varies minimum zero to maximum one. Larger the value indicates higher correlation. The detected firmness value has been applied to the rule based discriminant analyzer for classification of fruits as soft, medium and hard. Separation based on firmness gives only partial understanding about the fruit. Combination with auxiliary parameters can provide high grading accuracy.

III. RESULTS AND DISCUSSION

The ripening of mango fruit is directly related enzymatic breakdown which increases the fruit softness. The variations of firmness of matured fruit with number of days have been depicted in Figure-III. In the ripening stage the spectrometer data shows peak value in different spectrum values. The reflectance range 550–650 nm is associated with mango color changes and 750–1100 nm region linked with moisture and aromatic compounds.

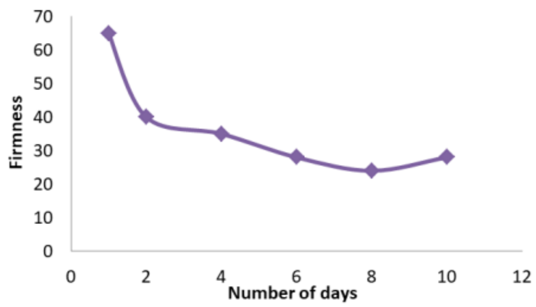

Figure-III Matured Mango firmness variation

Figure IV shows the variation of RMSECV with number of sampling. It decreases with the number of sampling and increases at the point of valid spectral information. It shows a minimum error at the 84th sampling interval.

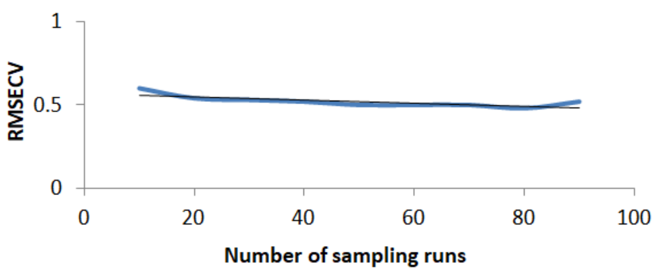

Figure IV Variation of RMSECV

The prediction correlation coefficient of the proposed model with Random Forest and combined methods can be provided in Table-II. It shows that the CARS+SVR model provides a high collinarity of 0.95 compared with existing methods.

Table-II Prediction results of proposed CARS+SVR Model

Extraction Method	Variable Number	R^2
CARS	32	0.912
Random Forest(RF)	12	0.862
SVR	4	0.842
CARS+RF	43	0.895
CARS+SVR	4	0.95

The mean squared prediction error comparison of proposed PLSR method with Principal Component Regression (PCR) has been illustrated in Figure V. The PLSR requires only two components to get same accuracy whereas PCR needs four components.

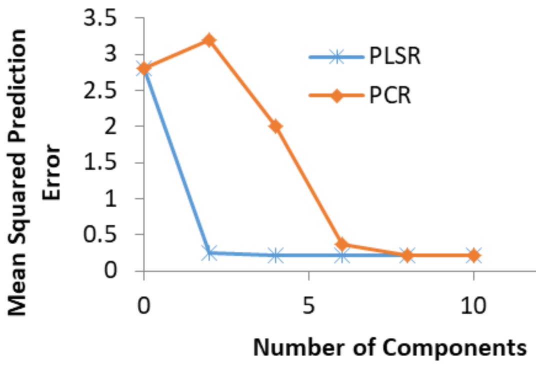

Figure V Mean squared Prediction Error Comparison

Table III projects the firmness prediction of three different mango varieties keitt, kent and alphonso in the spectral range of 900 to 1700 nm. The detected firmness has high prediction R^2value of 0.8 above for all varieties. As color has not been used as useful technique for firmness detection addition of non-destructive method can improvise the performance.

Table-III Determination of firmness for different cultivator

Cultivator	Spectral Range(nm)	Firmness(R^2)
Kent	950–1650	0.86
Keitt	684–990	0.84
Alphonso	900–1650	0.95

IV. Conclusion

The method provides a quick, reliable non-destructive visual near-infrared spectroscopic detection of mango pulp firmness developed with the hybrid framework of PLSR, CARS and SVR. The prediction model is developed spectral range of 900–1700 nm with fast image acquisition and performance measured by evaluation parameters such as correlation coefficient (R^2) and root mean square error (RMSEP). The dimensionality and over fitting of the dataset is greatly reduced with the help of hybrid prediction model. The results show that the proposed method can be used as a best tool to assist automatic grading in agricultural product quality maintenance field. In future the model performance has been extended to work in real time under different environment, light and temperature conditions. Addressing all these issues provides high accuracy and act as a reliable method for post-harvest management.

References

[1] Jahurul, M.H.; Zaidul, I.S.; Ghafoor, K.; Aljuhaimi, F.Y.; Nyam, K.L.; Norulaini, N.A.; Sahena, F.; Mohd, Omar, A.K. Mango (Mangifera indica L.) by-products and their valuable components: A review. Food Chem. 2015, 183, 173

[2] Sanchez, P. D. C., Hashim, N., Shamsudin, R., & Nor, M. Z. M. (2020). Applications of imaging and spectroscopy techniques for non-destructive quality evaluation of potatoes and sweet potatoes: A review. Trends in Food Science & Technology, 96, 208-221.

[3] Bundit Jarimopas, Udomsak Kitthawee, "Firmness Properties of Mangoes", International Journal of Food Properties, 10: 899–909, 2007

[4] S.K. Jha, "Firmness characteristics of mango hybrids under ambient storage", Journal of Food Engineering 97 (2010) 208–212

[5] Pathompong Penchaiya et al., "Measurement of Mango Firmness by Non-Destructive Limited Compression Technique", ISHS Acta Horticulturae 1088: II Southeast Asia Symposium on Quality Management in Postharvest Systems, 2013

[6] H.L. Wang, J.Y. Peng, C.Q. Xie, Y.D. Bao, Y. He, Fruit Quality Evaluation Using Spectroscopy Technology: A Review, Sensors 15 (2015) 118
.

[7] Arai, N.; Miyake, M.; Yamamoto, K.; Kajiwara, I.; Hosoya, N. Soft Mango Firmness Assessment Based on Rayleigh Waves Generated by a Laser-Induced Plasma Shock Wave Technique. Foods 2021, 10, 323. https://doi.org/10.3390/foods10020323

[8] Banlawe, I.A.P, Dela Cruz, J.C. Machine Learning-Based Classification of Mango Pulp Weevil Activity Utilizing an Acoustic Sensor, Micro machines 2023, 14, 1979. https://doi.org/10.3390/mi14111979

[9] Chaiwan, P.; Rachtanapun, P.; Phimolsiripol, Y.; Ruksiriwanich, W.; Li, C.; Luo, L.; Shen, D.; Chung, H.-H.; George, D.; Tangpao, T.; et al. Physicochemical Marker for Determination of Value-Adding Component in Over-Ripe Thai Mango Peels. Horticulturae 2024, 10, 1036. https://doi.org/10.3390/horticulturae10101036

[10] C. O'Brien et al., "Non-destructive methods for mango ripening prediction: Visible and near-infrared spectroscopy (visNIRS) and laser Doppler vibrometry (LDV)", Postharvest Biology and Technology,212,2024, https://doi.org/10.1016/j.postharvbio.2024.112878

[11] Kun Tao et al., "Improving the detection performance of mango firmness using a self-designed pneumatic-electromagnetic-driven impact device with the same impact force control ", International Journal of Agricultural & Biological Engineering, Vol. 18 No. 4,2025, https://www.ijabe.org

[12] Kamini G. Panchbhai & Madhusudan G. Lanjewar, "Identification of mango varieties with vitamin C and titratable acidity using stacking generalization from NIR spectra", Volume 19, pages 4257–4277, (2025).

[13] Walsh, K.B., Blasco, J., Zude-Sasse, M., Sun, X., 2020. Visible-NIR 'point' spectroscopy in postharvest fruit and vegetable assessment: The science behind three decades of commercial use. Postharvest Biol. Technol. 168, 1–17. https://doi.org/10.1016/j. postharvbio.2020.111246.

[14] Munawar, Agus Arip (2019), "Dataset of NIR spectrum for intact mango fruits", Mendeley Data, V1, doi: 10.17632/b9d6s7hr33.1

[15] Jha, S.N., Garg, R. Non-destructive prediction of quality of intact apple using near infrared spectroscopy. J Food Science Technology , 47, 207–213 (2010). https://doi.org/10.1007/s13197-010-0033-1

[16] Tang, G.; Huang, Y.; Tian, K.; Song, X.; Yan, H.; Hu, J.; Xiong, Y.; Min, S. A new spectral variable selection pattern using competitive adaptive reweighted sampling combined with successive projections algorithm. Analyst 2014, 139, 4894–4902.